ASMI BASNET

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PROFESSIONAL SUMMARY

BI Developer and Data Analytics Specialist based locally in Tallahassee, FL (Master of Science in Analytics candidate, Saint Louis University; B.E. in Electronics & Communication) with 3+ years of experience engineering enterprise BI dashboards, Kimball dimensional models, automated ETL workflows, and GIS geospatial intelligence solutions. Proficient in Power BI (DAX, Power Query, star-schema modeling), Tableau, advanced SQL (complex CTEs, window functions, query tuning), and GIS spatial integration (GeoJSON, Leaflet.js). Proven track record optimizing queries across 8+ databases to slash latency by 40%, architecting SCD Type-2 dimensional warehouses across 10,000+ records, and establishing automated data quality assertion frameworks (IV&V). Ready to support HNTB's digital infrastructure and transportation engineering projects across Florida.

TECHNICAL SKILLS & CORE COMPETENCIES

BI & Visualization: Microsoft Power BI (Advanced DAX, Power Query, Star/Snowflake Schemas, RLS, Incremental Refresh, Service), Tableau (LOD Expressions, Calculated Fields), Looker Studio, Excel (Power Pivot, VBA/Macros), Executive Reporting, KPI Design.
SQL & Data Engineering: Advanced SQL (Complex CTEs, Analytical Window Functions, Multi-Table Joins, Stored Procedures, Performance Tuning), PostgreSQL, SQL Server, Snowflake, DuckDB, BigQuery, SQLite, Python (Pandas, NumPy, SQLAlchemy), R.
GIS & Dimensional Modeling: GIS & Geospatial Data Integration, GeoJSON, Spatial Feeds & Highway Closure Telemetry, Leaflet.js, Kimball Dimensional Modeling, Star/Snowflake Schema Design, Slowly Changing Dimensions (SCD Type-2), 3NF Relational Schemas, Data Lineage.
Governance & Best Practices: Automated Data Quality Assurance (IV&V), Referential Integrity Checking, Schema Drift Detection, Metadata Repositories, Data Dictionaries, Technical Documentation, Unit Testing, Agile/Scrum Sprints, JIRA, Git / GitHub.

PROFESSIONAL EXPERIENCE

Saint Louis University — **St. Louis, MO**
Graduate Researcher, SPS Corps – Analytics Initiative — *August 2025 – Present*

- Architected enterprise data pipelines across 8+ operational databases, refactoring nested subqueries into modular SQL CTEs and analytical window functions (ROW_NUMBER, DENSE_RANK, LAG/LEAD) to reduce query latency by 40%.
- Built dynamic DAX measures, dimensional star-schema models, and interactive Power BI reports to present fiscal allocations, enrollment trends, and operational KPIs directly to university executive leadership.
- Implemented automated data quality verification suites (IV&V) that enforce referential integrity and detect schema drift, establishing centralized metadata repositories and comprehensive data dictionaries.
- Engineered automated ETL workflows and reconciliation checks, eliminating manual data entry cycles and accelerating weekly reporting turnarounds.
- Developed a hybrid forecasting engine pairing Prophet with PyTorch LSTM neural networks and automated anomaly detection (flagging variances beyond ±2.5 standard deviations) to provide leadership with predictive quarterly budget insights.
- Led requirements workshops with departmental stakeholders in Agile sprints to define reporting goals, authoring technical specifications, user stories, and conducting training sessions for 25+ staff members.

Saint Louis University — **St. Louis, MO**
Data Analyst Intern / BI Developer — *June 2024 – August 2024*

- Built and deployed 6 interactive Power BI and Tableau dashboards backed by star-schema data models, consolidating 15+ operational KPIs to enable leadership to monitor departmental productivity and project milestones.
- Authored and tuned SQL extraction queries across enterprise student databases using multi-table joins and window functions, reducing recurring reporting lag.
- Engineered automated data profiling and validation scripts in Python (Pandas) and SQL across 10,000+ institutional records, establishing webhook alerts that instantly flag data drift and transaction discrepancies.
- Enforced strict institutional data governance rules, establishing row-level security (RLS) and audit routines across sensitive reporting models.

Entegra Sources Pvt. Ltd. — **Kathmandu, Nepal (Remote)**
Data Analyst / Researcher & Systems Analyst — *December 2022 – July 2023*

- Extracted, cleansed, and mapped multi-source operational and telemetry datasets (5,000+ records) into standardized 3NF relational schemas using Python (Pandas, SQLAlchemy) and SQL, guaranteeing complete data fidelity.
- Validated algorithmic logic and automated backend processing scripts in Python and C/C++, implementing robust exception handling, boundary tests, and data reconciliation routines to maintain 99.8% pipeline reliability.
- Authored formal technical documentation detailing data dictionaries, validation rules, and sampling methodologies to establish organizational data governance standards.

FEATURED BI, GIS & DIMENSIONAL MODELING PROJECTS

Nepal Disaster Watch — Geospatial Intelligence & Real-Time Telemetry Platform — **TypeScript, GeoJSON, Leaflet.js, REST APIs**

- Architected a real-time GIS command platform integrating GeoJSON spatial feeds, live highway closures, river sensors, and meteorological data across all 77 districts of Nepal for OpenAI's WebMCP Challenge.
- Engineered resilient status adapters (live, snapshot, unreachable) and dynamic district cross-referencing tools, eliminating silent telemetry failures and enabling interactive spatial risk triage.

SCD Type-2 Dimensional Data Warehouse & Analytics Engine — **Python, SQL, DuckDB, Star Schema, Kimball Modeling**

- Designed a Kimball dimensional data warehouse with SCD Type-2 tracking (surrogate keys, effective/end dates, is_current flags) across 10,000+ records, preserving 100% point-in-time audit lineage.
- Authored complex SQL window functions (LAG, LEAD, DENSE_RANK) to calculate rolling retention, promotion velocity, and attrition metrics, feeding an interactive analytics dashboard.

CareGap — Clinical Decision Support & Population Health Analytics Engine — **Python, DuckDB, PostgreSQL, Power BI, Streamlit**

- Built a healthcare analytics data engine over 34,000+ longitudinal patient records, using columnar DuckDB for sub-second aggregations and PostgreSQL for transactional integrity.
- Delivered 6 interactive Power BI / Streamlit views tracking quality measures and care-gap closure rates, incorporating CDC growth-chart Z-score logic for pediatric cohorts.

EDUCATION & CERTIFICATIONS

Saint Louis University | Master of Science in Analytics (STEM) — **St. Louis, MO | Expected May 2026**
Tribhuvan University | Bachelor of Engineering in Electronics & Communication (STEM) — **Kathmandu, Nepal | Conferred 2023**
Certifications: AI Engineer for Data Scientists Associate (DataCamp) · Building with Claude & Anthropic API (Prompt Caching & Tools) · Project Management & Agile Certification (GlobalShala & SLU).